HAZBOT / HARDWARE INTERFACES AND NATIVE COMPUTE

Four subsystems around one Raspberry Pi

01 VISION / LOCAL ML

CSI ribbon

640 × 480 capture

Capture / inference on Pi

Pi Camera V2

Visual estimates: box / class / bearing

02 RANGING / NAVIGATION

Packets

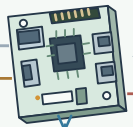
CP2102 USB–UART → USB

Built-in ranging MCU

YDLidar X4

RASPBERRY PI

Native inference + I/O services



RGB conversion / resize

INT8 TFLite detector

Boxes + class + confidence

Bearing / range association

Navigation / motion I/O

ROS 2 + serial services

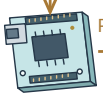
Scan parsing: angular range samples

Clearance and target-range inputs

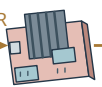
03 WHEEL DRIVE

Pi → drive MCU: direction commands

USB serial / 9600 baud



PWM / DIR



Power

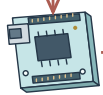
Drive Arduino **L298N driver** **DC wheel pair**

Separate low-level drive service; F / B / L / R / S

04 ARM / GRIPPER

Pi → arm MCU: joint commands

USB, 115200 baud



Servo pulses

Arm Arduino **3 axes + gripper**

B / S / A / G channels; dedicated D9–D12

Reference mission shown separately

Physical interface · Software data flow · Pi → MCU command direction

Schematic